

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Examination, 2016-17

Paper No. BSC-08A: Descriptive Statistics and Probability Distributions

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) What are the general rules of forming a frequency distribution? With the help of examples, explain the terms: (i) 'less than type' and 'greater than type' cumulative frequencies, (ii) width of a class interval, and (iii) mid-value of a class. [6]
- (b) Represent the following data by means of a histogram and cumulative frequency curve (ogive):

Weekly Wages (Rs)	10–15	15–20	20–25	25–30	30–40	40–60	60–80
No. of Workers	8	19	27	15	12	12	8

[8]

Question 2.

- (a) What do you mean by central tendency of the data? Define mean, median, and mode as measures of central tendency. [7]
- (b) Determine the missing frequency in the following incomplete distribution when its mode is 35:

Class Interval	0–10	10–20	20–30	30–40	40–50	50–60	60–70
Frequency	10	12	?	20	14	12	10

[7]

Question 3.

- (a) What do you mean by dispersion? Explain mean deviation and standard deviation with their relative merits and demerits. [7]
- (b) Define the coefficient of variation. The number of employees, average wages per employee, and variance of wages for two factories are given below. Find in which factory is there greater variation in the distribution of wages per employee:

	Factory A	Factory B
No. of Employees	100	200
Average Wages (Rs)	120	80
Variance of Wages	16	25

[7]

Question 4.

- (a) Define raw and central moments for grouped data. Calculate the first four central moments from the following set of raw data values:

2, 4, 6, 8, 10

[7]

- (b) What do you mean by skewness and kurtosis? The first four central moments of a set of data were found to be $\mu_1 = 0$, $\mu_2 = 9$, $\mu_3 = -4$, and $\mu_4 = 122$ respectively. What would you infer about the skewness and kurtosis of this data?

[7]

Question 5.

- (a) Define Karl Pearson's correlation coefficient and mention its main properties. [7]
- (b) Explain Spearman's rank correlation coefficient. Calculate Spearman's coefficient of correlation from the following data:

Rank by X	1	5	6	2	3	7	4	8
Rank by Y	3	7	8	1	2	6	5	4

[7]

Question 6.

- (a) Explain the use of scatter diagrams in studying the correlation between two variables. Plot the scatter diagram for the following data and comment:

X	1	2	3	4	5	6
Y	10	20	30	40	50	60

[7]

- (b) What do you mean by regression? Write down the expression for the two regression lines and show how the regression coefficients are related to the correlation coefficient. State their main properties. [7]

Question 7.

- (a) A six-faced ordinary die is thrown once. Find the probability of getting an outcome of:
(i) less than 3 spots, and (ii) more than 2 spots. [7]
- (b) State and prove the additive and multiplicative theorems of probability. [7]

Question 8. Write short notes on any **three** of the following:

- (a) Bar and pie diagrams
- (b) Normal distribution and its properties
- (c) Bayes' theorem
- (d) Probability mass function (p.m.f.) and probability density function (p.d.f.)

[14]